

# Toni Barth

Specialized practical employee at Anhalt University of Applied Sciences |  
Hochschule Anhalt

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Germany



My interest in digital media and programming was already apparent in my youth, which is why I invested a lot of time in learning various audio editing and programming techniques and also worked on individual open source projects.

At the beginning of my professional career, I then decided to do a dual degree in Applied Computer Science - Digital Media and Game Development in order to actively gain professional experience in addition to the basic theoretical knowledge. As part of this course, I was particularly interested in efficient programming with low-level languages such as C, but also in the development of websites, mobile apps or algorithms for artificial intelligence or multimedia signal processing. I have also worked with natural language processing and the efficient management of databases.

I also worked as a tutor for programming with C for fellow students in the 1st and 2nd semesters, and I was actively involved in the student council of my degree program and on the examination board to represent students at the university.

During my studies, I worked at GISA GmbH in Halle. Initially employed in the SAP environment, I switched to web development after 1 1/2 years, where I worked on the development of single-page apps based on AngularJS or Angular and the associated backend. I then also wrote my bachelor's thesis on the subject of upgrading the AngularJS web framework to the new Angular implementation in the business environment, which was awarded a grade of 1.0 in July 2018.

I completed my Master of Science from 2018 to 2021 at the Martin Luther University Halle-Wittenberg in cooperation with the Anhalt University of Applied Sciences in the Interactive Media department. I completed my studies with the master's thesis "Evaluation of the software environment Node.js as an alternative to Java in the development of enterprise software in the backend environment", which was awarded a grade of 1.0.

After completing my Master's degree, I moved to my alma mater, Anhalt University of Applied Sciences, to take up a position as a research assistant. In this role, I teach topics such as multimedia signal processing and web programming. I am also involved in the administration of Linux-based server systems.

Until 2020, I was part of the development team for the text-based game Avalon (available at <https://avalon.mud.de>) on a voluntary basis, which has been around for over 20 years and offers an incredible wealth of functions as well as challenges of both a technical and social nature.

I also regularly take on new tasks and discover new areas of interest for myself. Most of my projects are open source and can be found on GitHub under the username Timtam: <https://github.com/Timtam>

For example, you can also find all the results of the Global Game Jams there, which I have regularly participated in since 2016.

Since 2022, I have also been working as a consultant for accessibility, especially in audio applications. Thanks to my expertise in software development, I was able to work with Audient, for example, to make the control software for their iD and EVO interface series accessible: <https://audient.com/2023/08/14/accessibility-app/>

Since 2023 I have also been working part-time as a composer, sound designer, audio engineer and trainer, more information on my website <https://www.toni-barth.online/en>.

## Work Experience

### Specialized practical employee

[Anhalt University of Applied Sciences | Hochschule Anhalt](#) · Köthen (Anhalt), Saxony-Anhalt, Germany

July 2021 – present

## Volunteering

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### Community member and script author

REAPER accessibility community (German and English)

Ongoing support for blind and visually impaired REAPER users: answering questions, producing tutorials, maintaining accessibility-focused scripts, and hosting core community infrastructure.

- Maintaining a public ReaPack repository bundling accessibility scripts from the community: <https://github.com/Timtam/reapack>
- Hosting essential community services such as the REAPER Accessibility Hoard and the REAPER Accessibility Wiki

## Education

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### Master of Science - M.Sc., Interactive Media

Martin-Luther-University Halle-Wittenberg

April 2018 – June 2021

Final grade: 1,4

### Bachelor of Science - B.Sc., Applied Computer Science - Digital Media & Game Development

Anhalt University of Applied Sciences | Hochschule Anhalt

October 2014 – March 2018

Final grade: 1,6

## Awards

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### Best Paper Award (WEB 2026)

IARIA — International Academy, Research, and Industry Association

March 2026

Best Paper Award for "Towards Accessible Interactive Visual Elements on the Web: An AI-Based Descriptive and Auditory Input/Output Approach for Screen Reader Users" (with Leonie Weinert and Frank Heckel), presented at WEB 2026 in Valencia, Spain: [https://www.iaria.org/conferences2026/awardsWEB26/web2026\\_a2.pdf](https://www.iaria.org/conferences2026/awardsWEB26/web2026_a2.pdf)

### Köthener Informatikpreis

Anhalt University Of Applied Sciences | Hochschule Anhalt

October 2018

Award for the best bachelor thesis of the year

## Certificates

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### NVDA Certified Expert 2022

NV Access · [certification.nvaccess.org](https://certification.nvaccess.org)

July 2024

## Publications

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### Towards Accessible Interactive Visual Elements on the Web: An AI-Based Descriptive and Auditory Input/Output Approach for Screen Reader Users

WEB 2026, The Fourteenth International Conference on Building and Exploring Web Based Environments · [thinkmind.org](https://thinkmind.org)

March 2026

Interactive visual elements, such as digital whiteboards, have become essential components of modern collaboration, yet they remain largely inaccessible to screen reader users. While static images can be described using alternative text, dynamic and highly interactive visual content is particularly challenging due to rapid updates, spatial complexity, and the limitations of linear auditory output. This paper outlines a conceptual framework that leverages AI-driven event analysis, spatialized auditory feedback, and natural-language interaction to enable screen reader users to both perceive and manipulate interactive visual workspaces in real time. The proposed idea aims to close the accessibility gap by providing a bidirectional interaction loop that supports independent contribution in collaborative environments. Remaining challenges include efficient real-time change detection, minimizing cognitive load, and enabling privacy-preserving local computation. This work sets the foundation for future research toward more inclusive interaction paradigms for dynamic visual collaboration tools.

## Projects

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### ReaLackey

[github.com/Timtam/ReaLackey](https://github.com/Timtam/ReaLackey)

Native REAPER extension that integrates AI assistance directly into the DAW, letting users automate track, FX, MIDI and arrangement tasks through natural language. Built screen-reader-first (NVDA/OSARA), including screenshot-based interaction with otherwise inaccessible plugin interfaces.

*Rust, REAPER, AI, Accessibility, Screen readers*

### aperio

[timtam.github.io/aperio](https://timtam.github.io/aperio)

Accessible, keyboard-first calendar, tasks and contacts app for the desktop — screen-reader-first, local-first and provider-agnostic, connecting Google, iCloud/CalDAV, Outlook, Exchange, Todoist, Vikunja and local storage.

*Rust, Desktop, Productivity, Accessibility, Screen readers*

### Hitster online

[hitster.toni-barth.online](https://hitster.toni-barth.online)

Unofficial, fully accessible web implementation of the award-winning Hitster music card game, playable directly in the browser.

*TypeScript, Web, Games, Accessibility*

### Ear Dojo

[eardojo.com](https://eardojo.com)

Accessible music theory helpers and ear training in the browser (formerly Musical Sight), designed from the ground up to work well with screen reading technology.

*TypeScript, Web, Music theory, Ear training, Accessibility*

### Rabbit

[github.com/Timtam/rabbit](https://github.com/Timtam/rabbit)

Standalone tool that installs and updates REAPER, OSARA, SWS, ReaPack, ReaKontrol and more in one go — removing the biggest setup hurdle for blind and visually impaired REAPER users.

*Rust, REAPER, Accessibility, Tooling*

## Skills

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- Digital Accessibility: WCAG, ARIA, Screen reader testing (NVDA), Accessible audio workflows
- AI-Assisted Software Development: Agentic coding tools (Claude Code), Reviewing AI-generated changes, Used across recent projects such as ReaLackey
- AI Integration: LLM APIs (Claude, OpenAI, Gemini, Ollama), Natural-language interfaces, AI-based accessibility
- Interactive Media
- Algorithms
- Digital Signal Processing (DSP)
- Backend Development
- Video Games

## Languages

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- German: Native Speaker
- English: Fluent in written and spoken language